

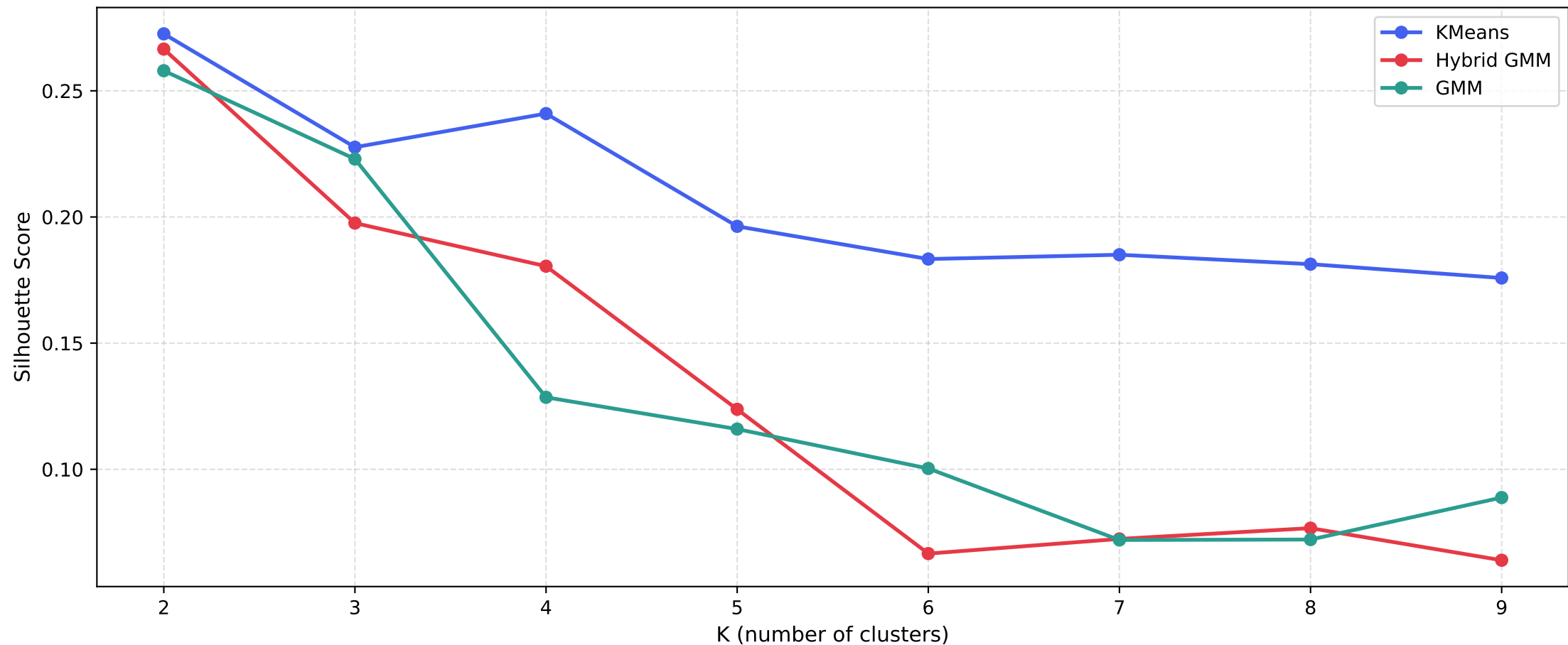
Clustering Model Comparison Report

KMeans · Hybrid GMM · GMM — UK Demographic Segmentation

1. Cluster Separation Metrics

Silhouette · Calinski-Harabasz · Davies-Bouldin

Silhouette Score by K

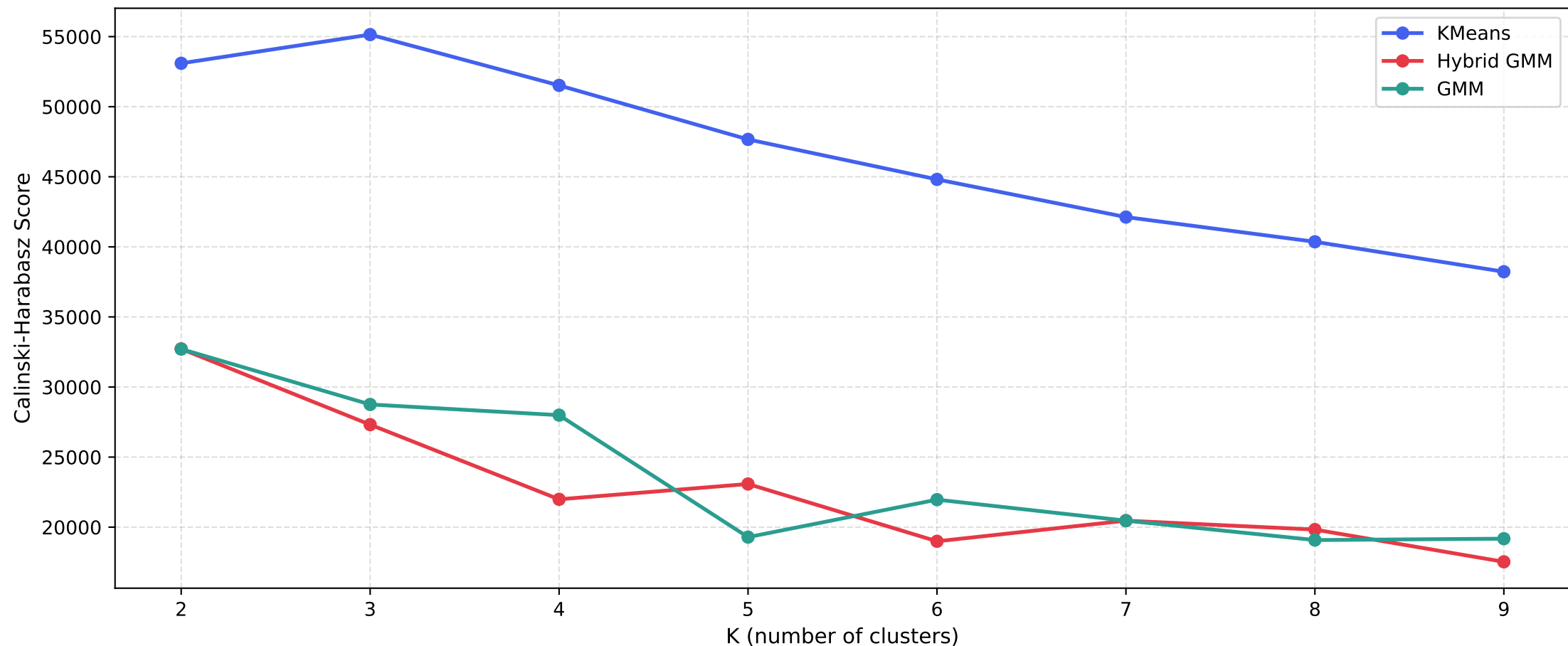


Measures mean intra-cluster cohesion vs inter-cluster separation. (↑ higher is better)

Silhouette Score — Data Table

k	KMeans	Hybrid GMM	GMM
2.0	0.2726	0.2665	0.258
3.0	0.2277	0.1976	0.223
4.0	0.241	0.1805	0.1285
5.0	0.1963	0.1238	0.1159
6.0	0.1833	0.0666	0.1003
7.0	0.185	0.0724	0.072
8.0	0.1813	0.0767	0.0722
9.0	0.1758	0.064	0.0888

Calinski-Harabasz Score by K

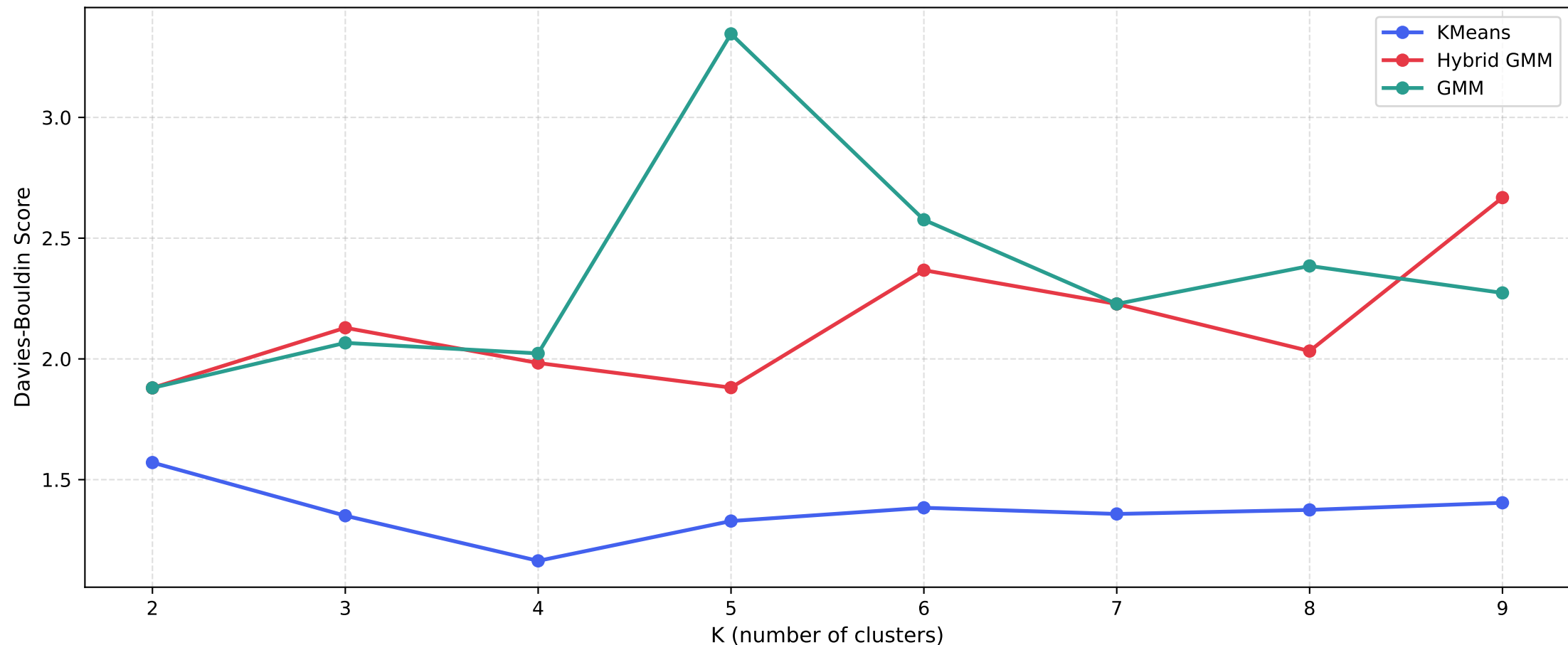


Ratio of between-cluster to within-cluster dispersion. Computed on full dataset. (↑ higher is better)

Calinski-Harabasz Score — Data Table

k	KMeans	Hybrid GMM	GMM
2.0	53097.481	32720.0733	32709.6785
3.0	55144.4072	27314.766	28754.4962
4.0	51520.722	21987.0669	27991.5042
5.0	47666.2326	23079.8431	19289.3289
6.0	44813.1371	18991.4495	21962.52
7.0	42124.694	20463.7695	20463.4638
8.0	40362.5314	19827.6088	19079.8276
9.0	38225.2311	17527.0068	19177.0241

Davies-Bouldin Score by K



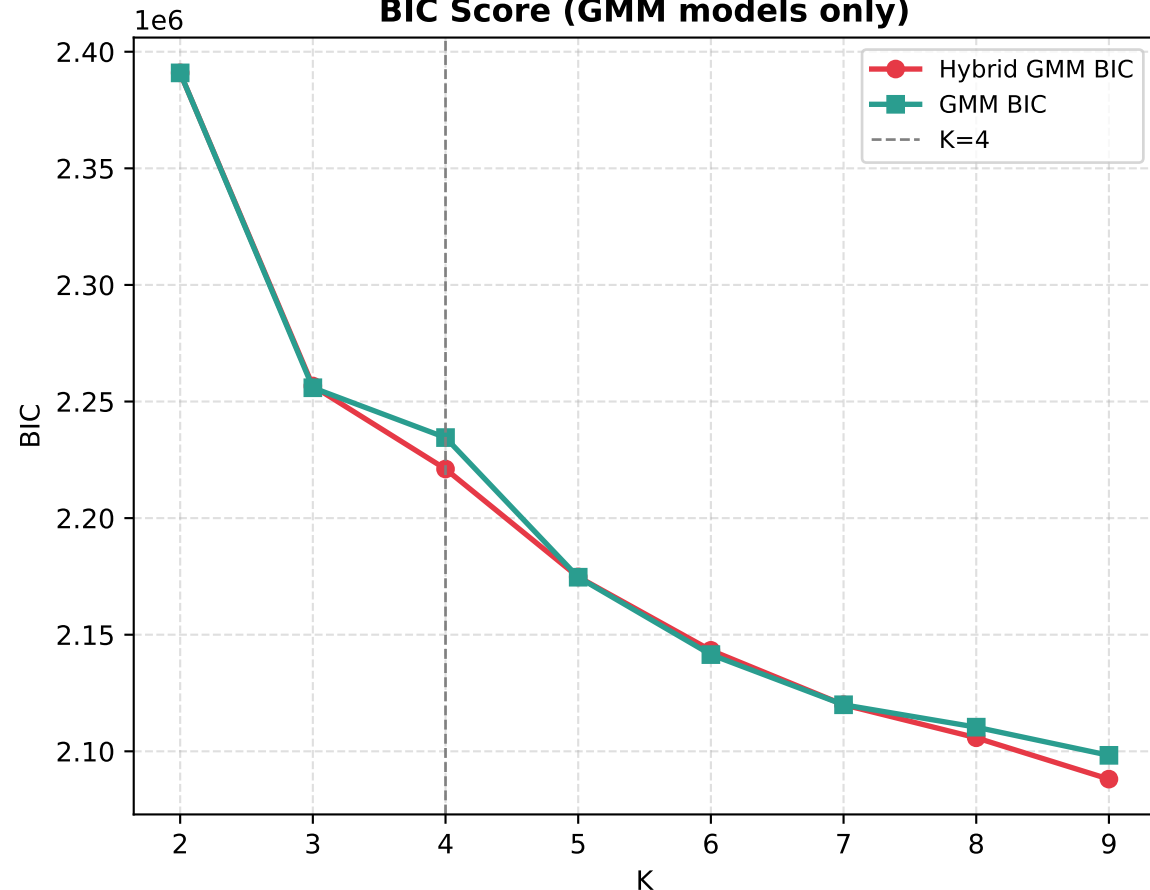
Average similarity between each cluster and its most similar neighbour. (↓ lower is better)

Davies-Bouldin Score — Data Table

k	KMeans	Hybrid GMM	GMM
2.0	1.5708	1.8799	1.88
3.0	1.3507	2.1289	2.0664
4.0	1.1638	1.9831	2.0228
5.0	1.3285	1.8811	3.3459
6.0	1.3836	2.3674	2.5761
7.0	1.358	2.2274	2.2276
8.0	1.3747	2.0325	2.3849
9.0	1.4042	2.6679	2.2732

2. Probabilistic Fit Metrics

BIC Score · Log-Likelihood (Hybrid GMM & GMM only)

BIC Score (GMM models only)

Lower BIC = better probabilistic fit penalised for complexity. KMeans excluded — no likelihood function.

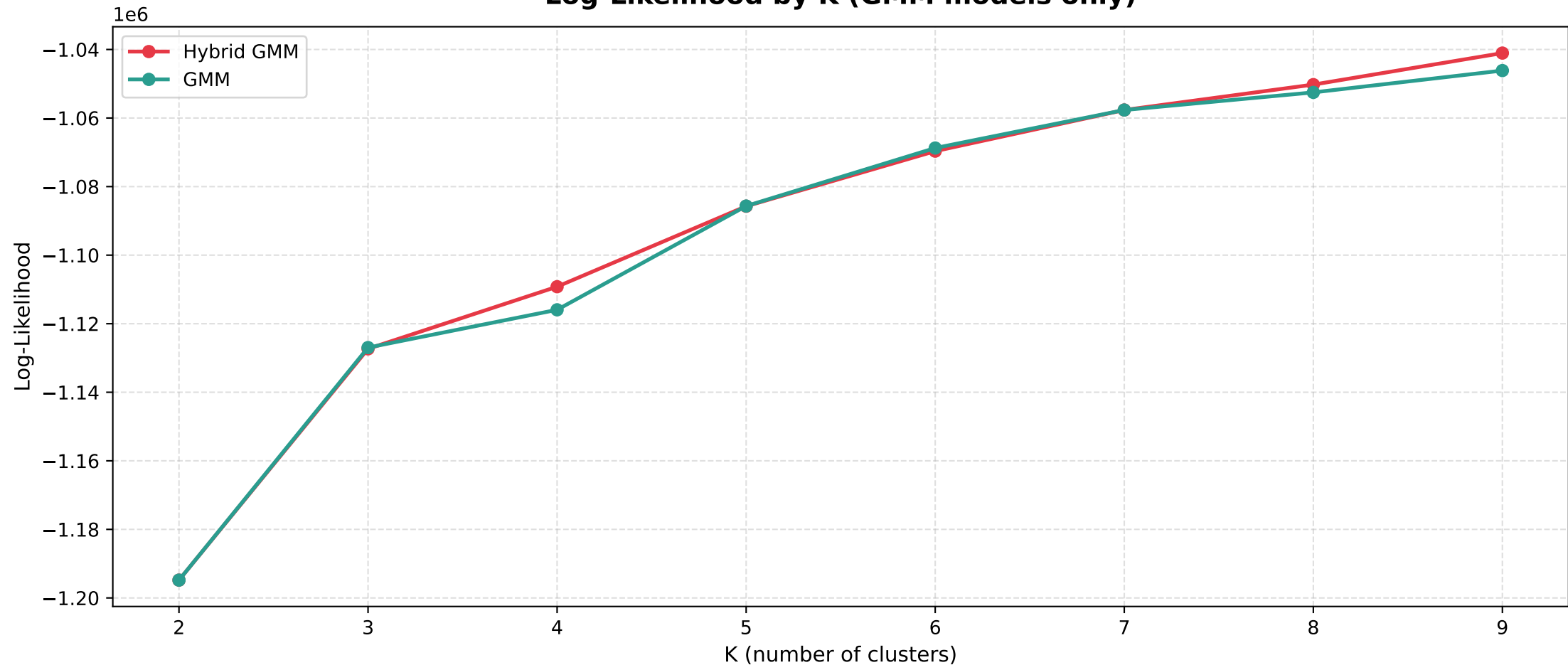
BIC Summary at K=4

Metric	Value
Hybrid GMM BIC at K=4	2,221,054
GMM BIC at K=4	2,234,540
Difference	+13,486
Penalty (%)	0.61%

BIC Score — Data Table

k	Hybrid GMM BIC	GMM BIC
2.0	2390934.9073	2390929.7736
3.0	2256645.9967	2255910.5204
4.0	2221053.9564	2234540.067
5.0	2174786.0211	2174658.2279
6.0	2143298.8891	2141473.1009
7.0	2119963.2104	2119959.5995
8.0	2105803.2469	2110351.2055
9.0	2088075.3453	2098275.4428

Log-Likelihood by K (GMM models only)

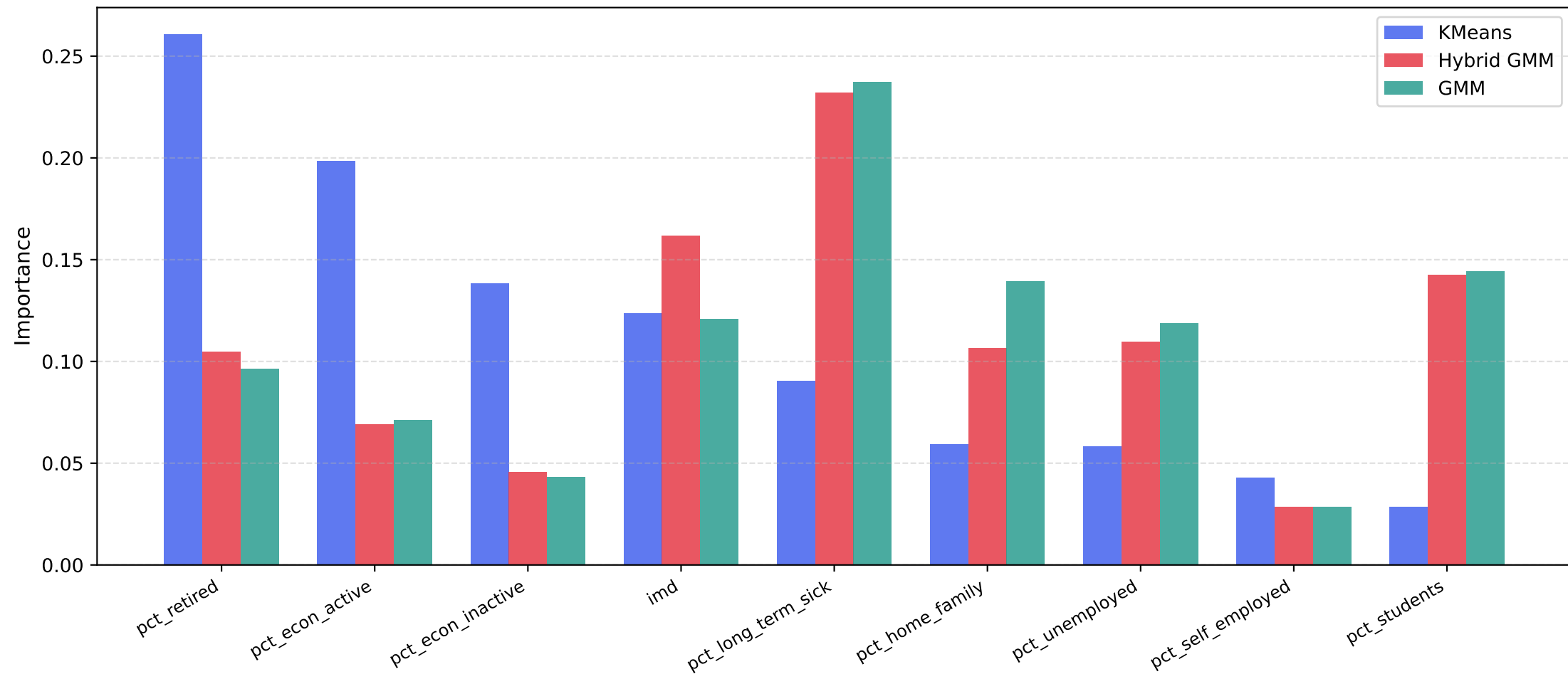


Higher (less negative) = better data fit. ($\uparrow$ higher is better)

3. Feature Importance

Random Forest Proxy — which Census variables drive segmentation?

Feature Importance by Model (Random Forest Proxy)



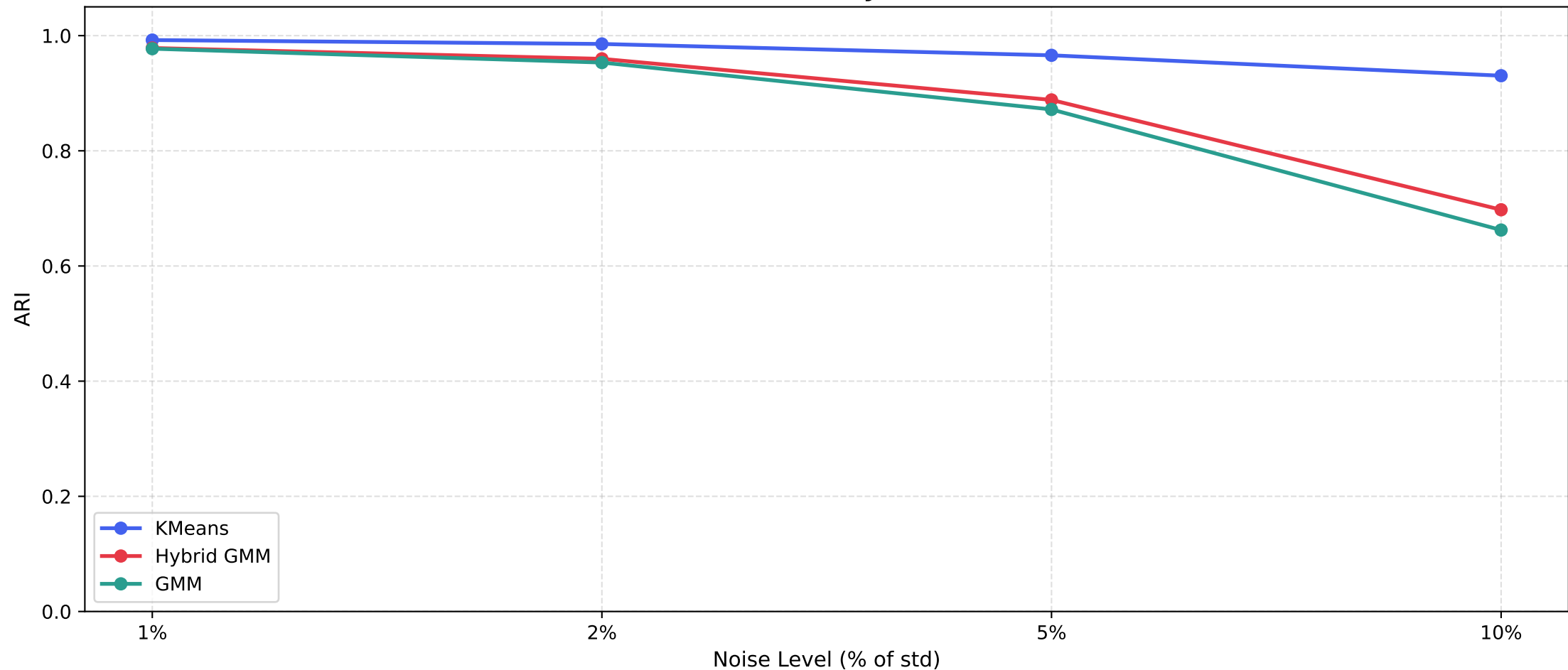
Feature Importance — Data Table

feature	KMeans	Hybrid GMM	GMM
pct_retired	0.2608	0.1046	0.0964
pct_econ_active	0.1983	0.0692	0.0713
pct_econ_inactive	0.1383	0.0455	0.0432
imd	0.1236	0.1619	0.1209
pct_long_term_sick	0.0903	0.2321	0.2373
pct_home_family	0.0591	0.1064	0.1393
pct_unemployed	0.0582	0.1095	0.1186
pct_self_employed	0.0428	0.0285	0.0285
pct_students	0.0285	0.1424	0.1444

4. Noise Robustness

Adjusted Rand Index under increasing Gaussian noise injection

Noise Robustness — Adjusted Rand Index



ARI = 1.0: identical assignments. ARI $\approx$ 0: random. Higher is better.

Noise Robustness — Data Table

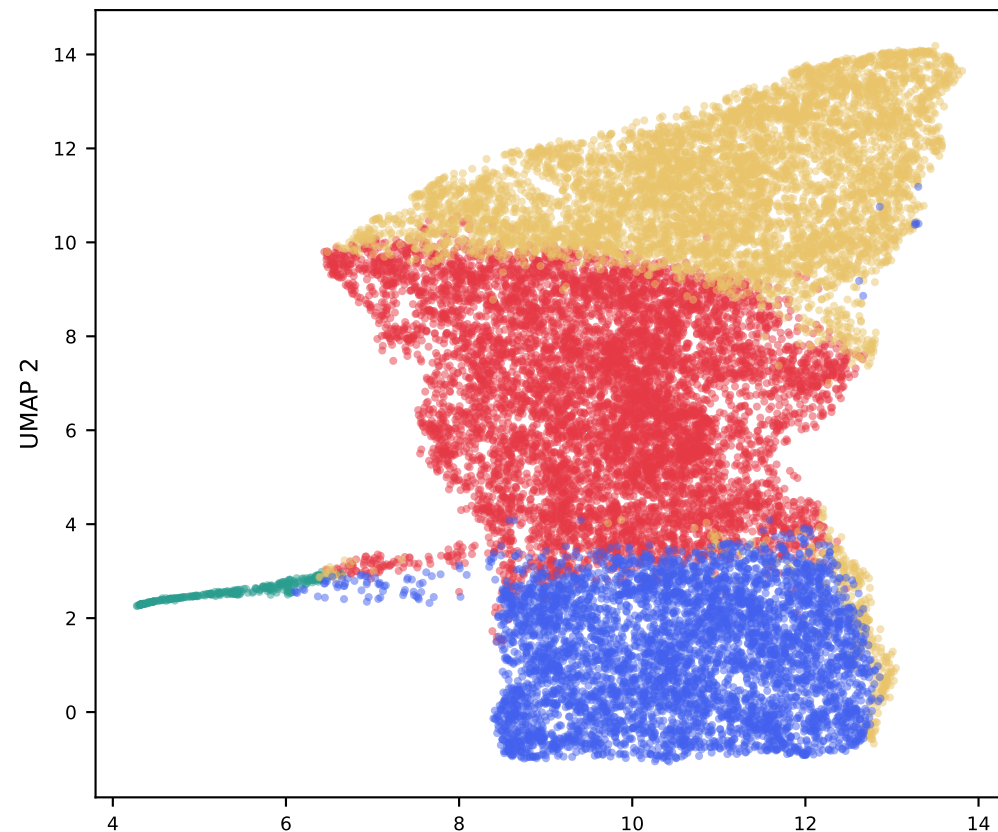
noise_level	KMeans	Hybrid GMM	GMM
1%	0.9923	0.9785	0.9773
2%	0.9855	0.9597	0.9533
5%	0.9658	0.8884	0.8718
10%	0.9304	0.6976	0.6624

5. UMAP Projections

2-D feature-space visualisation of cluster separation

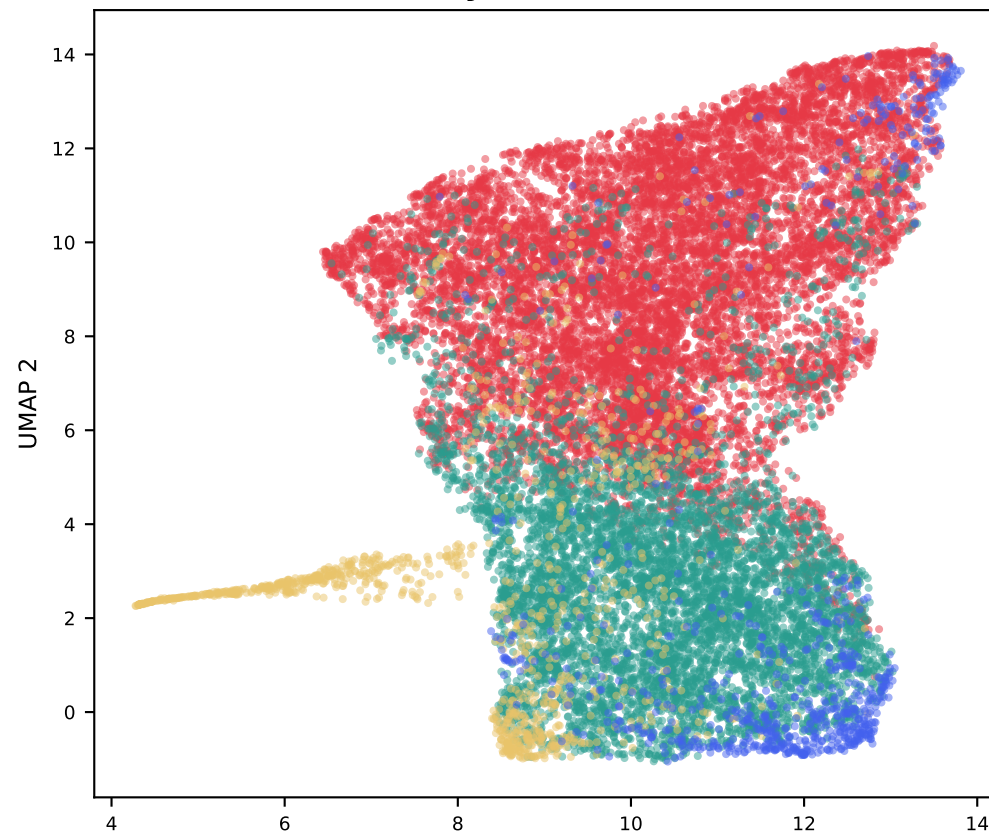
UMAP Projections — All Three Models

KMeans



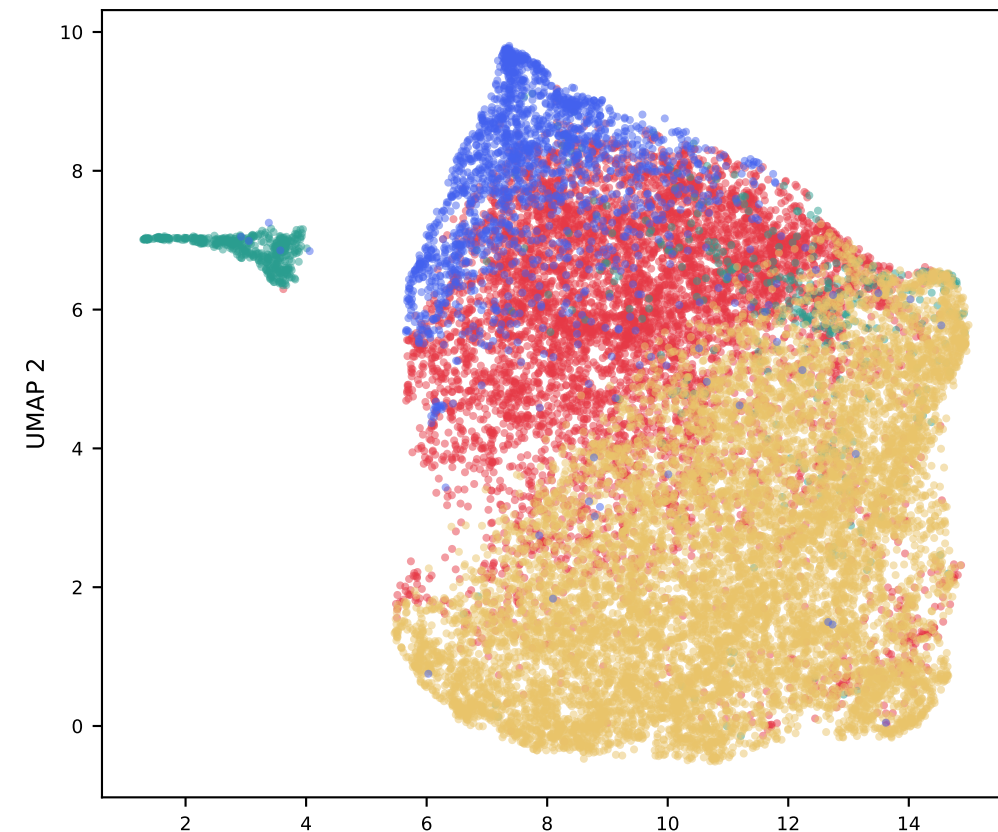
Rigid, spherical clusters.

Hybrid GMM

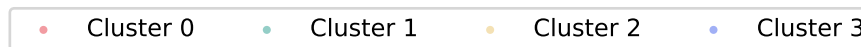


Probabilistic but often noisy.

GMM



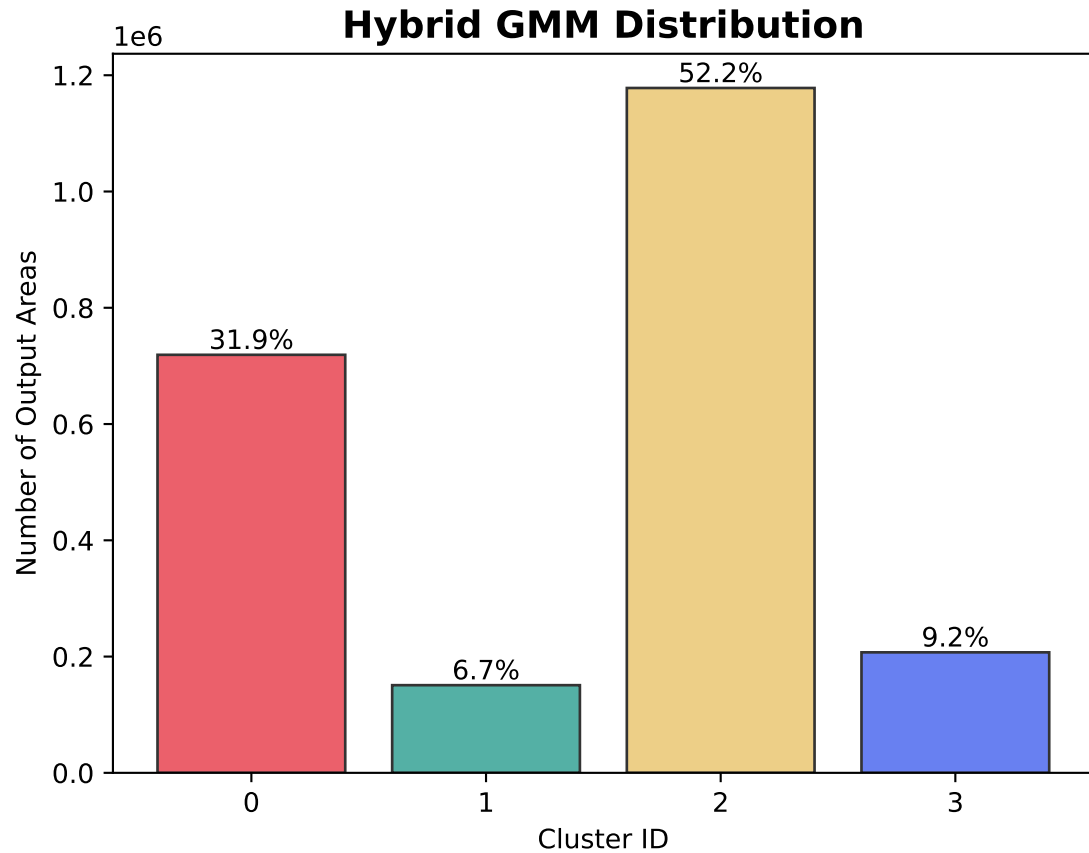
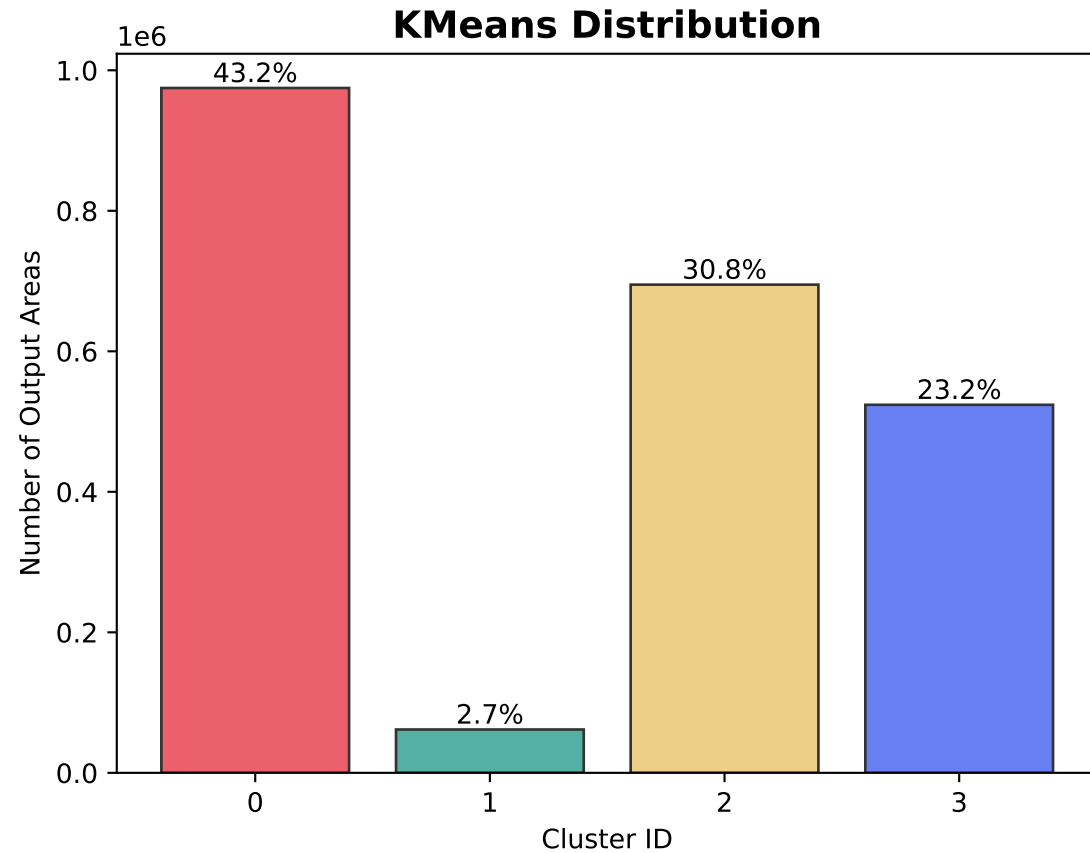
Refined boundaries with K-Means seeding.



6. Population Distribution

Comparing cluster volumes and balance

Volume Comparison: How models distribute the UK Population

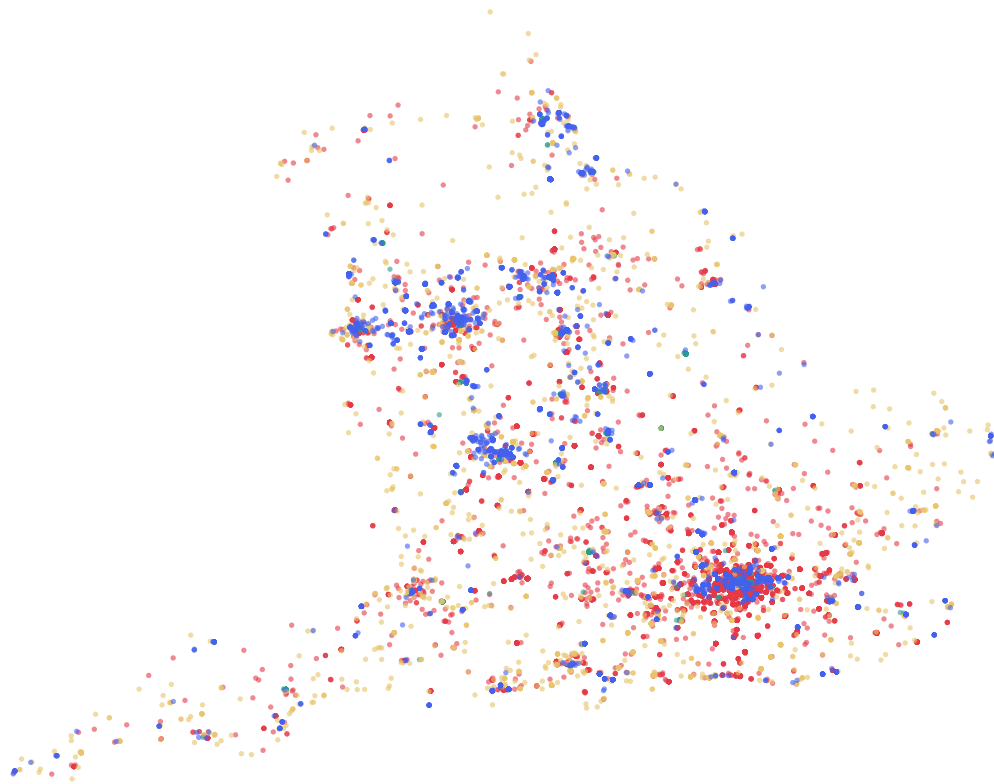


7. Geographic Footprint

UK-wide spatial distribution of segments

Geographic Footprint: KMeans vs. GMM Refinement

KMeans Map



GMM Map

